Finding a Hidden Point from Distance Queries on a Planar Region

Abstract

Let $S \subset \mathbb{R}^2$ be compact with positive area, and let P_s be a uniformly random point of S. A query is a point $P \in S$ and returns $\|P - P_s\|$. We prove that three queries always suffice to find P_s, and that two queries succeed with probability zero. Hence the answer is 3 for every such S.

1 Statement

Let $S \subset \mathbb{R}^2$ be compact with positive area, and let P_s be uniformly distributed on S. A strategy submits queries $P_1, P_2, \cdots \in S$, where P_k is a Borel function of $d_1, \ldots, d_{k-1}$ and $d_i = \|P_i - P_s\|$. The strategy finds P_s at step k if $P_k = P_s$.

Theorem 1. *For every such S:*
(a) some strategy finds P_s in at most three queries for every $P_s \in S$;
(b) every strategy finds P_s within two queries with probability 0.
Hence the minimum number of queries, counting the final one, is 3.

Notation. Let $H = \operatorname{conv} S$. It is compact, by Carathéodory's theorem, and convex, with nonempty interior because S has positive area. Let ∂H be its boundary. A segment is a nondegenerate segment. For distinct points P_1, P_2, let ℓ be the line through them and σ the reflection in ℓ.

2 Preliminary lemmas

Lemma 2 (Mirror pair)**.** *Let $P_1 \neq P_2$ and $d_i = \|P_i - P_s\|$. The points $x \in \mathbb{R}^2$ with $\|x - P_i\| = d_i$ for $i = 1, 2$ are exactly P_s and $\sigma(P_s)$.*

Proof. Choose coordinates with $P_1 = (0, 0)$, $P_2 = (D, 0)$, $D > 0$, so that ℓ is the first axis and $\sigma(x, y) = (x, -y)$. Subtracting $x^2 + y^2 = d_1^2$ and $(x - D)^2 + y^2 = d_2^2$ gives

$$x = a := \frac{d_1^2 - d_2^2 + D^2}{2D}, \qquad y = \pm h, \quad h := \sqrt{d_1^2 - a^2}.$$

Thus the solutions are $(a, \pm h)$. Since P_s is a solution, the solution set is $\{P_s, \sigma(P_s)\}$. □

Lemma 3 (Reduction)**.** *Let $P_1 \neq P_2$ in S. If $\sigma(P_s) \in S$, then $\sigma(P_s) \in H \cap \sigma(H)$. Consequently, if*

$$\sigma(P_s) \in H \cap \sigma(H) \implies \sigma(P_s) = P_s, \tag{1}$$

then P_s is the only point of S at distances d_1, d_2 from P_1, P_2, and a third query at that point finds P_s.

Proof. Since $P_s \in S$, we have $\sigma(P_s) \in \sigma(S) \subset \sigma(H)$; if also $\sigma(P_s) \in S$, then $\sigma(P_s) \in H$. By Lemma 2, a point of S at the given distances is P_s or $\sigma(P_s)$, and the latter lies in S only if it equals P_s by (1). □

Lemma 4 (Convex hull)**.** *(i) Every extreme point of H lies in S.*

(ii) If $p \in \partial H$ is not an extreme point of H, then ∂H contains a segment having p in its interior. Hence if ∂H contains no segment, then $\partial H \subset S$.

(iii) If ∂H contains a segment, then there are a supporting line ℓ of H and distinct extreme points A, B of H with $H \cap \ell = [A, B]$. In particular $A, B \in S$.

Proof. (i) Each point of H is a convex combination of finitely many points of S. An extreme point equals each point appearing with positive weight, so it lies in S.

(ii) Write $p = (x + y)/2$ with $x \neq y$ in H. Let f be a nonconstant affine function and c a constant with $f \leq c$ on H and $f(p) = c$, so $\ell = \{f = c\}$ supports H at p. From $f(x) + f(y) = 2c$ and $f \leq c$ we get $f(x) = f(y) = c$. Hence $[x, y] \subset H \cap \ell \subset \partial H$. The second statement follows from (i).

(iii) Let $I \subset \partial H$ be a segment with midpoint m, and let $\{f = c\} = \ell$ support H at m with $f \leq c$ on H. The affine function f is at most c on I and equals c at an interior point, so $f \equiv c$ on I and $I \subset \ell$. Thus $H \cap \ell$ is a compact convex subset of ℓ containing I, that is, a segment $[A, B]$. If $A = (x + y)/2$ with $x, y \in H$, then $f(x) = f(y) = c$ as in (ii), so $x, y \in [A, B]$ and therefore $x = y = A$. Thus A is extreme, and likewise B. By (i), $A, B \in S$. □

3 Upper bound

Proposition 5. *There is a strategy that finds P_s in at most three queries for every $P_s \in S$.*

Proof. By Lemma 3, it suffices to choose $P_1 \neq P_2$ in S such that (1) holds. If a query returns 0 we are done, so assume all distances are positive.

Case A: ∂H contains a segment. Take ℓ, A, B from Lemma 4(iii) and query $P_1 = A$, $P_2 = B$. Since ℓ supports H, the set H lies in one closed half-plane of ℓ and $\sigma(H)$ in the opposite one, so $H \cap \sigma(H) \subset \ell$. If $\sigma(P_s) \in H \cap \sigma(H)$, then $\sigma(P_s) \in \ell$, so $\sigma(P_s)$ is fixed by σ and

$$P_s = \sigma(\sigma(P_s)) = \sigma(P_s).$$

This is (1). Explicitly, let $D = \|B - A\|$, $u = (B - A)/D$, and n the unit normal of ℓ pointing into H. Then

$$P_s = A + a\,u + h\,n, \qquad a = \frac{d_1^2 - d_2^2 + D^2}{2D}, \quad h = \sqrt{d_1^2 - a^2} \ \ (\geq 0).$$

Case B: ∂H contains no segment. By Lemma 4(ii), $\partial H \subset S$. Query any $P_1 \in \partial H$ and read $d_1 > 0$.

For $P_2 \in \partial H \setminus \{P_1\}$, the open chord (P_1, P_2) misses ∂H, since otherwise ∂H would contain a segment by the proof of Lemma 4(ii). So the chord divides H into two convex pieces, each bounded by the chord and one of the two arcs of ∂H between P_1 and P_2. As $P_2 \to P_1$ along ∂H, one of these arcs shrinks to P_1. Choose P_2 so that one arc α satisfies

$$\delta := \operatorname{diam} \alpha < d_1.$$

Let ℓ^- be the closed half-plane of ℓ containing α, and let $K = H \cap \ell^-$ be the piece bounded by α and the chord. The extreme points of K lie on α, so $K = \operatorname{conv} \alpha$ and $\operatorname{diam} K = \delta$. Note that P_2 depends on d_1.

Claim. $H \cap \sigma(H) \subset K \cup \sigma(K)$. Let $x \in H \cap \sigma(H)$. If $x \in \ell^-$, then $x \in H \cap \ell^- = K$. Otherwise $\sigma(x) \in H \cap \ell^- = K$, so $x \in \sigma(K)$.

Now $P_1 \in K$ and $\sigma(P_1) = P_1$, so $K \cup \sigma(K)$ lies in the closed disc of radius δ about P_1. If $\sigma(P_s) \in H \cap \sigma(H)$, then

$$\|\sigma(P_s) - P_1\| \le \delta < d_1.$$

But $\|\sigma(P_s) - P_1\| = \|\sigma(P_s) - \sigma(P_1)\| = \|P_s - P_1\| = d_1$, a contradiction. Hence $\sigma(P_s) \notin H \cap \sigma(H)$ and (1) holds vacuously. □

Remark. No convexity or connectedness of S is assumed. The queries are extreme points of H, which lie in S even when the segment joining them does not.

4 Lower bound

Proposition 6. *For every strategy, the probability of finding P_s within two queries is* 0.

Proof. The first query P_1 is fixed, and $P_2 = g(d_1)$ for a Borel function g. The strategy succeeds at step 1 if $P_s = P_1$, an event of probability 0. It succeeds at step 2 exactly when $P_s \in E$, where

$$E = \{x \in S : \ x = g(\|x - P_1\|)\}$$

is a Borel set. For each $r > 0$, the circle of radius r about P_1 meets E in at most the one point $g(r)$. In polar coordinates about P_1,

$$\operatorname{area}(E) = \int_0^\infty \int_0^{2\pi} \mathbf{1}_E\big(P_1 + r(\cos\theta, \sin\theta)\big)\, d\theta\, r\, dr = 0,$$

because the inner integral is the measure of at most one point. Hence $\Pr[P_s \in E] = \operatorname{area}(E)/\operatorname{area}(S) = 0$. □

In words: the first distance places P_s on a circle, and the second query is a single point of it. If the strategy uses internal randomness independent of P_s, condition on it; the conclusion is unchanged.

Proof of Theorem 1. Part (a) is Proposition 5 and part (b) is Proposition 6. Since two queries succeed with probability 0, no strategy finds P_s in two queries for every $P_s \in S$, and three is the minimum. □